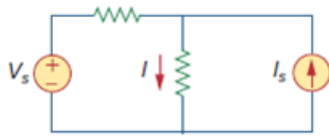


Chapter 4, Problem 9P

Problem

Given that $I = 6$ amps when $V_s = 160$ volts and $I_s = -10$ amps and $I = 5$ amp when $V_s = 200$ volts and $I_s = 0$, use superposition and linearity to determine the value of I when $V_s = 120$ volts and $I_s = 5$ amps.



Step-by-step solution

Step 1 of 12

Case: 1

Refer to Figure 4.77 in the textbook.

Redraw the circuit diagram when, $I = 6$ A, $V_s = 160$ V and $I_s = -10$ A.

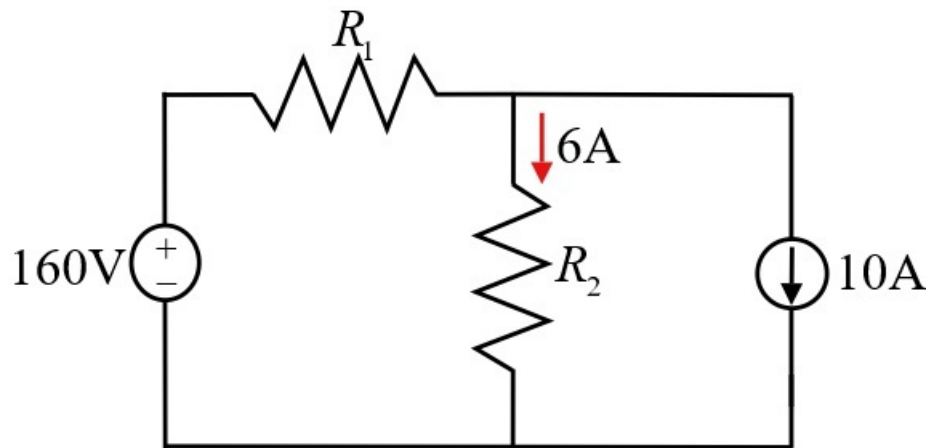


Figure 1

Step 2 of 12

Assume the current I is,

$$I = I_1 + I_2$$

$$6 = I_1 + I_2$$

$$I_1 = 6 - I_2$$

The current I_1 and I_2 are due to the voltage and current sources respectively.

Step 3 of 12

Apply the superposition theorem to Figure 1. Replace the current source by the open circuit.

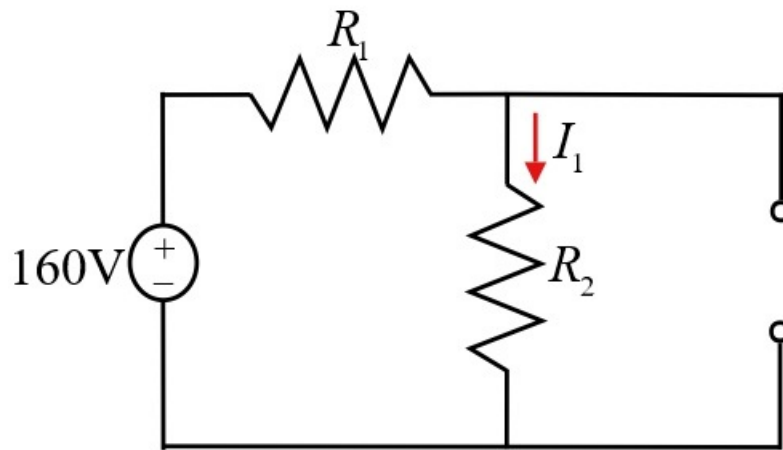


Figure 2

Step 4 of 12

Calculate the current I_1 .

$$I_1 = \frac{160}{R_1 + R_2}$$

$$R_1 + R_2 = \frac{160}{I_1}$$

Substitute $6 - I_2$ for I_1 .

$$R_1 + R_2 = \frac{160}{6 - I_2} \dots\dots (1)$$

Step 5 of 12

Replace the voltage source by the short circuit in Figure 1.

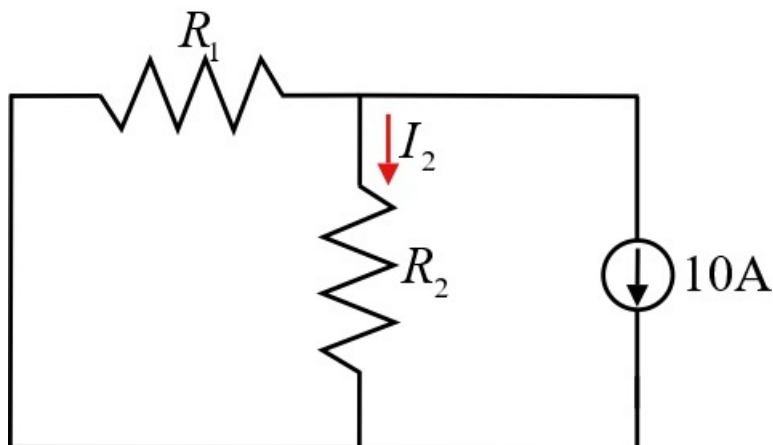


Figure 3

Step 6 of 12

Calculate the current I_2 in Figure 3.

$$I_2 = -10 \frac{R_1}{R_1 + R_2}$$

Substitute $-\frac{10R_1}{R_1 + R_2}$ for I_2 in equation (1).

$$R_1 + R_2 = \frac{160}{6 - \left(-\frac{10R_1}{R_1 + R_2} \right)}$$

$$R_1 + R_2 = \frac{160}{6 + \frac{10R_1}{R_1 + R_2}}$$

$$R_1 + R_2 = \frac{160(R_1 + R_2)}{16R_1 + 6R_2}$$

$$16R_1 + 6R_2 = 160$$

$$8R_1 + 3R_2 = 80 \dots\dots (2)$$

Step 7 of 12

Case: 2

Redraw the circuit diagram when, $I = 5 \text{ A}$, $V_s = 200 \text{ V}$ and $I_s = 0 \text{ A}$.

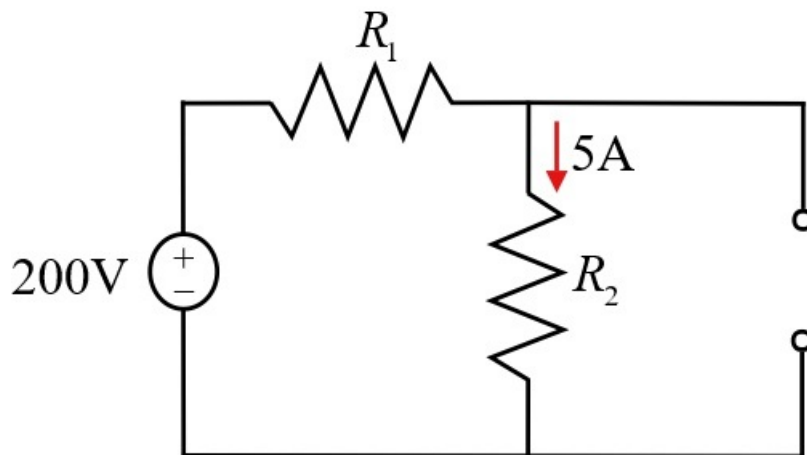


Figure 4

Step 8 of 12

Apply kirchhoff's voltage law in Figure 4.

$$200 = 5(R_1 + R_2)$$

$$R_1 + R_2 = 40 \dots\dots (3)$$

Solve equations (2) and (3).

$$R_1 = -8 \Omega$$

$$R_2 = 48 \Omega$$

Here, the negative value of the resistance is due to circuit is supplying power according to the passive sign convention.

Step 9 of 12

Redraw the Figure 1, when $I_s = 5\text{ A}$ and $V_s = 120\text{ V}$.

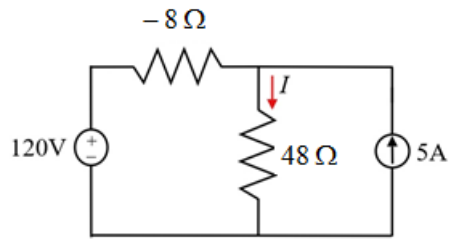


Figure 5

Step 10 of 12

Apply the superposition theorem to Figure 5 to find the current I . The current I is,

$$I = I_1 + I_2$$

Replace the current source by open circuit.

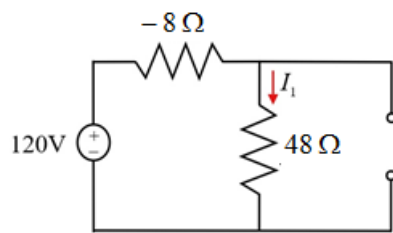


Figure 6

Step 11 of 12

Calculate the current I_1 .

$$\begin{aligned} I_1 &= \frac{120}{40} \\ &= 3\text{ A} \end{aligned}$$

Replace the voltage source by the short circuit.

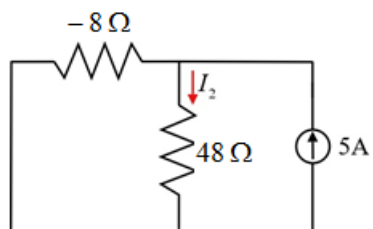


Figure 7

Step 12 of 12

Calculate the current I_2 .

$$\begin{aligned} I_2 &= \frac{-8 \times 5}{40} \\ &= -1 \text{ A} \end{aligned}$$

The total current I is,

$$\begin{aligned} I &= I_1 + I_2 \\ &= 3 - 1 \\ &= 2 \text{ A} \end{aligned}$$

Therefore, the value of current, I is 2 A.